Olist Delivery Performance — Business Requirements Document

Document Information

Project: Olist Delivery Performance Improvement Initiative
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1. Executive Summary

Olist, a Brazilian e-commerce marketplace, is experiencing measurable customer satisfaction and revenue loss due to late order deliveries. Analysis of ~96,000 delivered orders shows that 8.1% arrive after their promised delivery date, and late orders receive dramatically lower review scores (2.57 vs 4.29 stars). This document outlines the business problem, proposed solution, and requirements for a Delivery Risk Monitoring initiative to proactively identify and reduce late deliveries.

2. Problem Statement

Late deliveries are directly damaging customer trust and retention. Customers whose first order arrives late are significantly less likely to purchase again, and an estimated R$748,657 in revenue is tied to orders that are both late and poorly reviewed. The problem is not uniform — it concentrates in specific product categories (e.g., office_furniture) and geographic regions (notably Northeast Brazil states such as Alagoas and Maranhão), suggesting a targeted, rather than blanket, intervention is most effective.

3. Business Objectives

- Reduce the overall late-delivery rate from 8.1% to below 5% within 2 quarters
- Improve average review score for previously late-prone categories/regions
- Reduce revenue loss associated with late + poorly-reviewed orders by 25%
- Improve first-order-to-repeat-purchase conversion rate

4. Scope

In Scope:

- Identification of high-risk orders (by category and region) before dispatch
- Proactive alerting to operations/logistics teams for high-risk shipments
- Monthly reporting dashboard on delivery performance by category and state

Out of Scope:

- Changes to Olist's underlying shipping carrier contracts
- Real-time GPS tracking implementation
- Customer-facing delivery promise changes (future phase)

5. Stakeholders

Role	Interest
VP of Operations	Owns delivery performance KPIs
Logistics Managers	Execute day-to-day shipping decisions
Customer Experience Team	Owns review scores and retention
Data/Analytics Team	Builds and maintains the risk model and dashboard
Category Managers	Accountable for category-level seller performance

6. Current State Process

Today, orders are shipped with no proactive flagging of delivery risk. Delays are discovered only after the fact, through customer complaints or post-delivery review scores. There is no mechanism connecting historical category/region late-rates to new incoming orders.

7. Proposed Solution

Introduce a Delivery Risk Score, calculated per order at the point of sale using historical late-rate data by product category and customer state. Orders flagged as high-risk trigger an alert to logistics for expedited handling or proactive customer communication about possible delays.

8. Functional Requirements

- FR1: System shall calculate a delivery risk score for each order based on category and state historical late-rate
- FR2: System shall flag orders above a defined risk threshold (e.g., >12% historical late rate)
- FR3: System shall send an alert to the logistics team for flagged orders
- FR4: System shall provide a dashboard showing late-rate trends by category and state, refreshed weekly
- FR5: System shall allow category managers to view seller-level performance within their category

9. Non-Functional Requirements

- NFR1: Dashboard must load within 5 seconds for standard date ranges
- NFR2: Risk scoring must run within the existing order-processing window (no added latency to checkout)
- NFR3: All customer and order data must remain compliant with Brazil's LGPD data protection law

10. Success Metrics / KPIs

- Late delivery rate (target: <5%)
- Average review score for flagged-category orders
- Revenue tied to late + bad-review orders (target: 25% reduction)
- Repeat purchase rate for customers with high-risk first orders

11. Assumptions & Constraints

- Historical category/state late-rate patterns are assumed to persist in the near term
- Solution depends on existing order and logistics data being available in real time
- No additional headcount assumed for initial rollout

12. User Stories

Epic: Delivery Risk Monitoring

- As a Logistics Manager, I want to see which incoming orders are flagged as high delivery-risk, so that I can prioritize them for expedited handling.
- As a Logistics Manager, I want an automatic alert when a high-risk order is placed, so that I don't have to manually check every order against historical performance data.
- As a Category Manager, I want to see late-delivery rates broken down by seller within my category, so that I can hold underperforming sellers accountable.
- As a VP of Operations, I want a weekly dashboard showing late-delivery trends by category and state, so that I can track whether interventions are working.
- As a Customer Experience Lead, I want to know which customers just received a late first order, so that I can proactively reach out and reduce churn risk.
- As a Data Analyst, I want the risk score calculation to be based on rolling historical late-rate data, so that the model stays accurate as patterns shift over time.

Acceptance Criteria (example - Logistics Alert story):

- Given an order is placed in a category/state combination with a historical late rate above 12%, when the order is confirmed, then the system flags it as "High Risk" and notifies the logistics team within 5 minutes.
- Given a flagged order, when the logistics team views it, then they see the order's category, state, and historical late rate as context.